EDUAI ACADEMIC SMART NOTES

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University Curriculum Series

# Introduction to Machine Learning

## Section Topic: Supervised Learning: Regression & Classification

**Executive Summary:** Deep dive into regression and classification mathematics, gradient descent derivation, tree ensembles, and robust model evaluation metrics.

## 1. Core Learning Objectives

- Master Ordinary Least Squares (OLS) and Gradient Descent optimization for Linear Regression
- Formulate Logistic Regression, the Sigmoid activation, and Binary Cross-Entropy loss
- Compare margin-based classifiers (Support Vector Machines) with tree-based ensembles
- Evaluate multi-class classification using Confusion Matrices, Precision, Recall, and ROC-AUC

## 2. Conceptual Breakdown & Technical Formulations

### 1. Linear Regression and the Cost Surface

Linear regression models the relationship between continuous dependent scalar y and independent explanatory variables X.

The hypothesis function is:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d = \theta^T x$$

We formulate the Mean Squared Error (MSE) objective function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

To find parameters $\theta$ that minimize $J(\theta)$, we employ **Gradient Descent**:

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) = \theta_j - \frac{\alpha}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

Where $\alpha > 0$ is the learning rate hyperparameter.

### 2. Logistic Regression & Binary Cross-Entropy

For classification, predicting continuous numbers is ill-suited. We map raw logits $z = \theta^T x$ into the interval $(0, 1)$ via the **Logistic Sigmoid function**:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Interpreting $\hat{p} = \sigma(z) = P(y=1|x)$, we maximize log-likelihood, leading to the **Binary Cross-Entropy (BCE) Loss**:

\mathcal{L}_{BCE}(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[ y^{(i)} \ln(\hat{p}^{(i)}) + (1 - y^{(i)}) \ln(1 - \hat{p}^{(i)}) \right]

### 3. Decision Trees and Ensembles (Random Forest, Gradient Boosting)

Decision Trees recursively partition feature space by choosing splits that maximize **Information Gain** (minimizing Entropy or Gini Impurity):

\text{Entropy}(S) = - \sum_{k=1}^C p_k \log_2(p_k)

\text{Gini}(S) = 1 - \sum_{k=1}^C p_k^2

- **Random Forest**: Bagging (Bootstrap Aggregating) + random feature subspacing to reduce variance.
- **Gradient Boosting (XGBoost/LightGBM)**: Sequential boosting where each tree fits the negative gradient residuals of previous trees.

### 4. Comprehensive Evaluation Metrics

Accuracy alone is misleading on imbalanced datasets. We define:

- **Precision**: \frac{TP}{TP + FP} (Fraction of positive predictions that were correct)
- **Recall (Sensitivity)**: \frac{TP}{TP + FN} (Fraction of actual positives captured)
- **F1-Score**: Harmonic mean: 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}
- **ROC-AUC**: Area under the True Positive Rate vs False Positive Rate curve across all classification thresholds.

### 5. Practical Scikit-Learn Code

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, confusion_matrix
rf = RandomForestClassifier(n_estimators=100, max_depth=6, random_state=42)
rf.fit(X_train, y_train)
y_pred = rf.predict(X_test)
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

### 6. Exam-Important Takeaways

- Always inspect the cost function convexity: Linear & Logistic regression have convex loss functions (guaranteed global minimum under gradient descent).
- When false negatives are critical (e.g., cancer detection), optimize for **Recall**.
- When false positives are disastrous (e.g., fraud blockings), prioritize **Precision**.

## 3. Self-Assessment & Exam Checklist

| Topic Milestone | Mastery Check |
|---|---|
| Can you formulate the primary mathematical loss / model objective? | [ ] Confirmed |
| Can you distinguish parametric vs non-parametric assumptions? | [ ] Confirmed |

| | |
|---|---|
| Are you prepared to calculate precision, recall, and F1 metrics? | [ ] Confirmed |
| Do you understand the difference between L1 sparsity and L2 shrinkage? | [ ] Confirmed |